

SECTION 38 — PROGRESS TRACKER & MASTER INDEX

Purpose: Executive status dashboard and topic coverage index for the Universal Enterprise Senior SDET & Staff QA Architect Preparation Notebook (Tier-1 Tech, FAANG, Big 4, Enterprise Cloud).

Master Section Index & Status Table

Section	Title	File Name	Status	Line Count	Key Topics Covered
0	Interview Strategy	SECTION_0__INTERVIEW_STRATEGY.md	✓ Complete	~367	Competency Matrix, STAR format, Strategy
1	Core Java Foundations	SECTION_1__CORE_JAVA_FOUNDATIONS.md	✓ Complete	~608	JVM, JRE, Bytecode, Stack/Heap, G1GC
2	Object-Oriented Programming	SECTION_2__OBJECTORIENTED_PROGRAMMING.md	✓ Complete	~590	Encapsulation, SOLID, Dynamic Dispatch
3	Advanced Java Concepts	SECTION_3__ADVANCED_JAVA_CONCEPTS.md	✓ Complete	~480	Equals/HashCode, Reflection, Immutability, String Masterclass
4	Java Collections Framework	SECTION_4__JAVA_COLLECTIONS_FRAMEWORK.md	✓ Complete	~324	List, Set, Queue, Concurrent Collections, Utility Tables
5	HashMap & Hash-Based Collections	SECTION_5__HASHMAP_AND_HASHBASED_COLLECTIONS.md	✓ Complete	~291	Bins, Treeification, Load Factor, CHM
6	Generics	SECTION_6__GENERICS.md	✓ Complete	~275	Wildcards, Type Erasure, Type Bounds
7	Exceptions & Error Handling	SECTION_7__EXCEPTIONS_AND_ERROR_HANDLING.md	✓ Complete	~230	Checked/Unchecked, Try-with-resources
8	Java 8 Features	SECTION_8__JAVA_8_FEATURES.md	✓ Complete	~280	Streams, Lambdas, Optional & Collectors Masterclass
9	Multithreading & Concurrency	SECTION_9__MULTITHREADING_AND_CONCURRENCY.md	✓ Complete	~326	ThreadPools, Atomic, Synchronization
10	Java Coding & DSA for SDET	SECTION_10__JAVA_CODING_AND_DSA_FOR_SDET.md	✓ Complete	~408	Arrays, Strings, Two-Pointer, Sliding Window, Regex Parsing
11	Selenium WebDriver Architecture	SECTION_11__SELENIUM_WEBDRIVER_ARCHITECTURE.md	✓ Complete	~287	W3C Protocol, Drivers, CDP Bridge
12	Selenium Locators & Strategy	SECTION_12__SELENIUM_LOCATORS_CODE_AND_STRATEGY.md	✓ Complete	~242	Relative XPath, Dynamic Locators

Section	Title	File Name	Status	Line Count	Key Topics Covered
13	Selenium Element Interactions	SECTION_13__SELENIUM_ELEMENT_INTERACTIONS.md	✓ Complete	~251	Actions, Frames, Windows, Alerts
14	Selenium Synchronization & Waits	SECTION_14__SELENIUM_SYNCHRONIZATION_AND_WAITS.md	✓ Complete	~234	FluentWait, Custom ExpectedConditions
15	Page Object Model & Design	SECTION_15__SELENIUM_PAGE_OBJECT_MODEL_AND_FRAMEWORK_DESIGN.md	✓ Complete	~1428	Layered POM, LoadableComponent
16	Selenium Advanced Scenarios	SECTION_16__SELENIUM_ADVANCED_SCENARIOS.md	✓ Complete	~1177	Grid, Dynamic Data, DevTools Protocol
17	TestNG Framework	SECTION_17__TESTNG__COMPLETE_INTERVIEW_SECTION.md	✓ Complete	~784	Parallel Execution, Listeners, DataProvider
18	REST Assured & API Automation	SECTION_18__REST_ASSURED_AND_API_AUTOMATION.md	✓ Complete	~2046	POJO Serialization, BDD, Schema Validation
19	Playwright with TypeScript	SECTION_19__PLAYWRIGHT_WITH_TYPESCRIPT.md	✓ Complete	~1249	Contexts, Auto-wait, Web Locators
20	Test Automation Architecture	SECTION_20__TEST_AUTOMATION_ARCHITECTURE.md	✓ Complete	~1469	Framework Layers, Multi-tenant Design
21	CI/CD for Senior SDET	SECTION_21__CICD_FOR_SENIOR_SDET.md	✓ Complete	~653	Pipelines, Matrix Sharding, Quality Gates
22	Parallel Testing & Scalability	SECTION_22__PARALLEL_TESTING_AND_SCALABILITY.md	✓ Complete	~237	Thread Safety, Worker Isolation
23	Test Data Management	SECTION_23__TEST_DATA_MANAGEMENT.md	✓ Complete	~193	Data Factories, Ephemeral Provisioning
24	Database & SQL for SDET	SECTION_24__DATABASE__SQL_FOR_SDET.md	✓ Complete	~338	Window Functions, Joins, Execution Order
25	API & System Testing Scenarios	SECTION_25__API_TESTING_AND_SYSTEM_TESTING_SCENARIOS.md	✓ Complete	~216	Microservices, Contract Testing (Pact)
26	Performance & Reliability	SECTION_26__PERFORMANCE__RELIABILITY_AWARENESS.md	✓ Complete	~172	k6, Latency Percentiles (p95/p99), Load Profiles
27	Testing Strategy & QA Fundamentals	SECTION_27__TESTING_STRATEGY_AND_QA_FUNDAMENTALS.md	✓ Complete	~216	Shift-Left, Risk-based Testing
28	Debugging & Failure Analysis	SECTION_28__DEBUGGING_AND_FAILURE_ANALYSIS.md	✓ Complete	~229	Root Cause Triage, Trace Viewer
29	Flaky Test Management	SECTION_29__FLAKY_TEST_MANAGEMENT.md	✓ Complete	~155	Quarantine Engine, Flake SLA, Root Cause
30	Framework Code Snippet Library	SECTION_30__FRAMEWORK_CODE_SNIPPET_LIBRARY.md	✓ Complete	~464	Reusable Drivers, Specs, Utilities

Section	Title	File Name	Status	Line Count	Key Topics Covered
31	Design Patterns for Test Automation	SECTION_31__DESIGN_PATTERNS_FOR_TEST_AUTOMATION.md	Complete	~209	Builder, Factory, Strategy, Singleton
32	System Design for SDET	SECTION_32__SYSTEM_DESIGN_FOR_SDET.md	Complete	~114	Enterprise Grid, Observability Stack
33	Behavioral & Leadership	SECTION_33__BEHAVIORAL_AND_LEADERSHIP.md	Complete	~131	Universal Core Values, 4 STAR+ Stories
34	Project Deep-Dive	SECTION_34__PROJECT_DEEP_DIVE.md	Complete	~283	Enterprise Cloud Modernization & Banking Suite
35	Rapid-Fire Interview Questions	SECTION_35__RAPIDFIRE_INTERVIEW_QUESTIONS.md	Complete	~1032	160+ Q&A Pairs + Quick Reference
36	Mock Interview Sets	SECTION_36__MOCK_INTERVIEW_SETS.md	Complete	~483	10 Structural Mocks + Scoring Rubric
37	Final Revision Sheets	SECTION_37__FINAL_REVISION_SHEETS.md	Complete	~1807	10 One-Page Sheets + 24h Checklist
38	Progress Tracker	SECTION_38__PROGRESS_TRACKER.md	Complete	~100	Master Section Index & Status Dashboard

Core Readiness Verification Checklist

- ✔ **Java / JVM Mechanics:** Garbage collection, ThreadLocal memory leaks, HashMap internals, String Masterclass & Immutability.
- ✔ **Playwright & TypeScript:** BrowserContext, auto-waiting, custom fixtures, storageState authentication.
- ✔ **API & REST Assured:** Jackson serialization, JSON Schema validation, dynamic payloads.
- ✔ **Framework & Systems Architecture:** ThreadLocal isolation, dynamic data factories, config management.
- ✔ **CI/CD & DevOps:** GitHub Actions matrix sharding, blob report merging, deployment canary strategy.
- ✔ **SQL Mastery:** Query execution order, window functions (`DENSE_RANK` , `LAG`), anti-joins.
- ✔ **Interview Performance:** STAR leadership stories, 60-min universal mock interview rubric, 24-hour rapid revision checklist.

38.3 Code & Governance Reference

```
# Automated Notebook Audit Command
python3 audit_notebook.py --verify-all
```

38.4 Anti-Patterns & Pitfalls

- **Skipping Weak Topics:** Focusing only on UI automation while ignoring JVM memory architecture and SQL window functions.